

Augmented Humans

Cognitive, physical and agential extensions of the human body



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May 2026 · Version 1.0

DOI: pending Zenodo assignment

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ABSTRACT

This white paper defines the **augmented human** as a person who incorporates, in a persistent and non-accessory manner, technological extensions that modify their cognitive, sensory, motor or agential capacity. Four convergent vectors of augmentation are analysed: brain-computer interfaces (BCIs) from companies such as **Neuralink, Synchron and Blackrock Neurotech**; medical and industrial exoskeletons; permanent personal AI agents; and sensory neural prostheses. The central philosophical question of the period —how far does a human remain human when a growing share of their functions is assisted or executed by integrated technology?— is examined, and a five-principle ethical framework is proposed. The paper argues that Latin America holds the conditions to constitute itself as a regional hub of the field, provided it consolidates a regulatory framework, public investment and specific clinical training over the next five years.

Keywords: Augmented human · Brain-computer interfaces · Neuralink · Synchron · Exoskeletons · Permanent personal agents · Bioethics · Neural prostheses · LATAM hub

“An augmented human is not a better human. It is a different human, who makes different decisions, assumes different responsibilities and must be conceived by a law that does not yet exist.”

— Chris Meniw

1. Introduction

The idea of augmenting the human body is ancient: spectacles, hearing aids, pacemakers and joint prostheses form part of a historical continuity that long precedes the contemporary conversation. What changes in the present, and demands a new analytical category, is the *active, persistent and connected* nature of available extensions. A pacemaker sustains a function; a brain-computer interface *amplifies* one. A hearing aid restores; a permanent personal AI agent *adds*.

I have observed over the last decade that public debate moves between two sterile extremes: transhumanist utopianism celebrating any augmentation, and bio-essentialist conservatism rejecting any modification. This white paper proposes a rigorous middle path, both technically and ethically sustainable.

2. Canonical definition

An **augmented human** is defined as a person who incorporates, in a persistent fashion —understood as use exceeding fifty percent of active hours— one or more technological extensions that measurably modify their cognitive, sensory, motor or agential capacity relative to the biological baseline.

The definition deliberately excludes episodic tools (a search engine consulted three times a day does not constitute augmentation) and includes external systems whenever integration is structural and functional dependence is real. The criterion is not invasiveness but *operational integration*.

3. Vector I: brain-computer interfaces

BCIs constitute the most spectacular vector and, simultaneously, the earliest in maturity. Three players define the horizon:

Neuralink operates on a dense intracortical electrode paradigm, with published cases of cursor and device control in paraplegic patients. Iteration is fast; bioethical controversy is equally so.

Synchron develops the stentrode, a less invasive endovascular interface implantable without open cranial surgery. Clinical cases in patients with ALS and other motor-neuron pathologies.

Blackrock Neurotech is the actor with the longest academic track record: its Utah array has operated in human clinical settings for more than fifteen years and constitutes the methodological reference of the field.

The three lines converge on a common question: when does a BCI stop being a restorative medical device and become an elective augmentation device. The answer is not technical; it is regulatory.

4. Vector II: exoskeletons

Exoskeletons divide into two lines: **medical** (motor rehabilitation, assistance for spinal-injury patients) and **industrial** (fatigue reduction in logistics, construction, elderly care). The second segment grows at double-digit rates per year and is silently transforming entire trades.

Unlike BCIs, exoskeletons do not provoke significant cultural rejection: society reads them as tools, not modifications. Prolonged use, however, raises serious questions about muscle atrophy, functional

dependence and occupational liability when the device fails.

5. Vector III: permanent personal AI agents

The least visible and, paradoxically, the most widespread vector. A permanent personal AI agent is a system that accompanies the person continuously, with access to their context, memory and preferences, and acts as a cognitive extension. It requires no implant nor exclusive wearable hardware; it runs on everyday devices.

In operational terms, a professional with a permanent personal agent operates with extended cognitive capacities: they remember everything the agent remembers, synthesise at the speed of the agent, access bodies of knowledge that without the agent would take months to traverse. **ZOE IA**, in its corporate deployments, exemplifies this category: permanent agents that accompany professional teams throughout their full working day, not as occasional tools.

6. Vector IV: sensory neural prostheses

Next-generation cochlear implants, retinal prostheses and vestibular-stimulation systems constitute a quiet but clinically consolidated field of augmentation. Their consolidation opens the question of the limit: when a cochlear prosthesis not only restores hearing but extends it to frequencies inaudible to the biological ear, are we facing treatment or augmentation?

7. The central philosophical question

Ethical conversation matures when it stops asking “should it be allowed?” and starts asking “how is it distributed?”. I have argued in various forums that the real question of the period is not whether humans will augment themselves—they already are— but under what regime of access, accountability and legal identity.

I propose five minimum principles: (i) structural and revocable informed consent, (ii) mandatory interoperability between vendors—no human should remain captive to a company because their implant is not compatible with another’s—, (iii) right to reversal when technically possible, (iv) special protection of the data generated by augmentation (neural, biometric, cognitive-pattern data), and (v) guaranteed access to a minimum clinically justified augmentation, regardless of socio-economic level.

8. Implications and LATAM opportunity

Latin America holds three favourable conditions to constitute itself as a regional hub of the field: (i) a high-quality clinical base concentrated in five to seven reference medical centres, (ii) regulatory frameworks more agile than European ones for responsible clinical research, and (iii) comparative costs that make regional development of components and software viable.

The unfavourable conditions are equally clear: regulatory dispersion across countries, low public investment in neurotechnology, talent exodus. The **Chris Meniw Foundation Inc.** drives a regional agenda articulated on three axes: minimum bioethical harmonisation, regional clinical-research fund, and scientific-talent retention programme. If the decade is seized, Latin America can move from importer of neurotechnology to recognised producer. General conclusion: the augmented human is

not future, it is present; what is being decided now is what kind of augmented human we are going to be.

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How to cite this whitepaper (APA)

Meniw, C. (2026). *Augmented Humans: Cognitive, physical and agential extensions of the human body*. Chris Meniw Foundation Inc. <https://doi.org/10.5281/zenodo.XXXXX>